

TEJAS PANDYA

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EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING , Brooklyn, NY	Expected 05/27
Master of Science in Financial Engineering GPA: 3.8/4.0	
Graduate Teaching Assistant: Financial Economics; C++ Financial Software Laboratory; Value Investing	
SVKM'S NMIMS, SCHOOL OF BUSINESS MANAGEMENT , Mumbai, India	04/22
Master of Business Administration, Concentration in Finance GPA: 3.3/4.0	
VELLORE INSTITUTE OF TECHNOLOGY , Vellore, India	06/20
Bachelor of Technology in Mechanical Engineering GPA: 8.9/10.0	

PROGRAMMING / TECHNICAL SKILLS / CERTIFICATIONS

Programming: Python, C/C++ (C++20, STL), SQL, Bash, LaTeX/Typst
Libraries & Frameworks: NumPy, Pandas, Polars, SciPy, scikit-learn, statsmodels, LightGBM, XGBoost, PyTorch, pytest
Data & Tools: Linux/Unix, Git, CMake, Streamlit, Shiny, Excel, Bloomberg Terminal, Tableau, Power BI
Certifications: CFA Level I; C++ Programming for Financial Engineering (Baruch College); Bloomberg Market Concepts; CFI FMVA

COURSEWORK HIGHLIGHTS

Quantitative Methods & Computing: Financial Econometrics, Machine Learning in Financial Engineering, Quantitative Methods in Finance, Financial Computing, Applied Numerical Methods, Data Structures and Algorithms
Risk, Valuation & Derivatives: Financial Risk Management, Valuation for Financial Engineering, Quantitative Portfolio Management, Options Pricing & Stochastic Calculus, Fixed Income Securities and Interest Rate Derivatives

EXPERIENCE

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING , Brooklyn, NY	05/26 - 08/26
<i>Quantitative Researcher, QWIM Industry Capstone</i>	
- Lifted Sharpe from 0.37 to 0.55 against a 60/40 benchmark over 1999-2025 as one of five on the QWIM capstone, pairing equal-risk-contribution weighting with hierarchical budgeting on Conditional Drawdown-at-Risk and blending regime-specific covariance by one-month-ahead probability - Implemented a log-periodic power-law singularity bubble-confidence tilt as the fourth ablation level above regime-conditioned covariance blending and quantile-regression tail estimation, the strongest of eight variants, holding Sharpe 0.56 against 0.50 on the 2008-2025 window - Built the Regime Risk Budgeting dashboard tab and its Typst client-report exporter, 1,058 lines of Python across ten modules with twelve unit tests, written fail-safe so that any missing artifact degrades only that one report section rather than breaking the whole client report	
NOMURA SERVICES INDIA PRIVATE LIMITED , Mumbai, India	08/22 - 07/25
<i>Senior Risk Analyst, Market Risk Infrastructure</i>	
- Owned the lifecycle of VaR and sensitivity models across equities, rates, FX and commodities, from daily production runs through backtesting; Python anomaly detection triaged exceptions to stale marks, erroneous data or booking errors, cutting investigation turnaround 40% and restoring SLAs - Delivered quarterly Basel 2.5 and FRTB stress-testing and capital submissions alongside front-office and regulatory teams, engineering rules-based validation over delta, vega and curvature sensitivities that reconciled fully against front-office pricing and caught anomalies pre-submission - Engineered Python and SQL pipelines across equities, rates, FX and commodities that aggregate, validate and reconcile daily P&L, positions and Level-1/Level-2 market data into the VaR and sensitivity dashboards, eliminating more than 50 hours of manual processing every month - Led a three-person workstream on US-region end-of-day risk reporting, cutting overnight batch runtime 35% through SQL query optimisation and job scheduling, and built the VaR, P&L and exception dashboards that senior stakeholders read each morning before the US open	

ACADEMIC PROJECTS

CROSS-SECTIONAL EQUITY ALPHA RESEARCH	07/26
<i>Python, LightGBM, statsmodels</i> github.com/tejaspandya9598/equity-xs-alpha	
- Constructed eight cross-sectional signals across 495 S&P 500 names, scored by Spearman rank information coefficient with Newey-West t-statistics under Benjamini-Hochberg false-discovery control; only Amihud illiquidity survived at 5.32, while the low-volatility and lottery signals inverted - Scored 15,575 EDGAR filings by Loughran-McDonald tone and traced that factor's 1.37 t-statistic to a 10-K versus 10-Q calendar artefact; differencing within form collapsed it to 0.46, while a LightGBM composite reached 0.79 Sharpe against -0.57 on a survivorship-biased panel	
MACRO-REGIME TACTICAL ASSET ALLOCATION	12/25 - 07/26
<i>Python, cvxpy, MOSEK, scikit-learn</i> github.com/tejaspandya9598/macro-regime-allocation	
- Detected macro regimes from 120+ FRED-MD indicators by PCA to 95% of variance into two-step K-Means, then benchmarked that detector against Wasserstein K-medoids, a Gaussian HMM and NBER dating across 798 months, finding it switches every 3.8 months against the 44.7 NBER takes - Found a look-ahead in the published backtest, with regimes fitted on 1959-2023 data labelling every month traded, and measured it by refitting the detector yearly: the headline Ridge book fell from Sharpe 0.65 to 0.35, below equal weight, while the regime-mean book rose to 0.59	
ALMGREN-CHRISS OPTIMAL EXECUTION	05/26
<i>Python, NumPy, Deribit API</i> github.com/tejaspandya9598/optimal-execution	
- Implemented the closed-form Almgren-Chriss liquidation schedule minimising expected cost plus lambda times variance, calibrated to live Deribit BTC-PERPETUAL volatility and average daily volume; it takes about 38% off TWAP's cost standard deviation, not a lower expected cost - Trained a NumPy policy-gradient agent with hand-derived gradients checked against central differences in continuous integration, on 207,361 real five-minute bars; it converged to within 0.060bps of the closed form it was never shown, and nothing beat TWAP out of sample	